UNIT 4 · 9 hours

Common Source Amplifier



Transistors

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

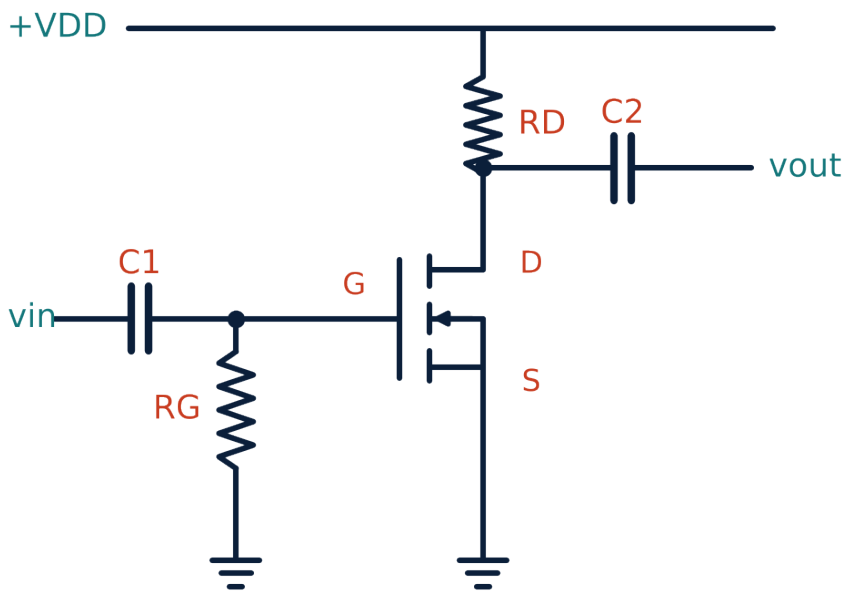
- Draw the common-source (CS) amplifier with a MOSFET
- Explain biasing and the role of each component
- Define transconductance and voltage gain
- Show that the CS stage inverts the signal

Syllabus context

Construction and operation, BJT as a switch, working and characteristics of n-channel enhancement type MOSFET, common source amplifier, DC load line analysis, MOS inverter.

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CS amplifier circuit



- Input at the gate (via C1), output at the drain (via C2).
- Source is common to input and output (grounded).
- RG sets the gate DC level; RD is the drain load.
- MOSFET operated in saturation.

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How amplification occurs

1

Bias

DC gate voltage sets ID and Q-point in saturation.

2

Signal

A small vin changes VGS by vgs.

3

Current changes

id = gm vgs, where gm = ∂ID/∂VGS.

4

Output

vout = −id RD: amplified and inverted.

Rising input → rising ID → larger drop across RD → falling drain voltage: 180° phase shift.

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Small-signal gain

$$g_m = 2k\,(V_{GS} - V_T)$$

transconductance (S)

$$A_v = v_{out}/v_{in} = -\,g_m R_D$$

voltage gain (neglect ro)

$$A_v = -\,g_m\,(R_D \parallel R_L)$$

with load RL

- Negative sign: 180° phase inversion.
- Higher gm or RD gives higher gain; gain limited by saturation of VDS.

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Numerical: gain of a CS stage

Problem A MOSFET with $k = 0.5\ mA/V^2$ is biased at $V_{GS} - V_T = 2\ V$. $R_D = 5\ k\Omega$. Find g_m and A_v.

- $g_m = 2k(V_{GS} - V_T) = 2 \times 0.5\ mA/V^2 \times 2\ V = 2\ mA/V$
- $A_v = -g_m R_D = -2\ mS \times 5\ k\Omega$

Answer: $g_m = 2\ mA/V$, $A_v = -10$

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Components and features

C1, C2: coupling capacitors block DC, pass the signal.

RG: very large (MΩ) so the high input resistance is preserved.

Features: high input impedance, moderate-high voltage gain, 180° phase shift.

Uses: voltage amplifier stage in analog ICs, preamplifiers.

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Key takeaways

- CS amplifier: input at gate, output at drain, source common
- gm = 2k(VGS − VT); Av = −gm RD
- Output is inverted (180° phase shift)
- Bias in saturation for linear operation

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Practice questions

Q1. Draw a CS amplifier and label all components.

Q2. Find gain if gm = 3 mS and RD = 4.7 kΩ.

Q3. Why does the CS stage invert?

Q4. What is the function of the coupling capacitors?